

I am Y — a clean restatement

Difference, loop, observer and coherence in a protected space

Original author: E R A Craig, New Zealand. **This version:** a rewriting of "*I am Y but I don't know what or where I am, I feel in the dark with what I haven't*" in which every sentence is intended to be defensible as written. The structure, the vocabulary and the order of the original are kept; the section numbers correspond one-to-one. What changed is the language: each statement now carries a label,

[def] a definition · [stip] a stipulated choice · [thm] a proved statement, with the name of its machine-checked proof · [open] a question this document does not settle,

and no sentence claims more than its label allows. The original text is untouched and remains in the project; the audit that produced these labels is `Y_OBSERVER_STUDY_REPORT.md`; the proofs are in `RequestProject/ObserverY.lean`; the arithmetic is reproduced exactly by `observer_y.py --selftest`.

0. Position of this document

This document does not claim that reality is this system.

It claims something narrower and checkable:

Given a stipulated read cost, a stipulated zone cost and a chosen protected geometry, there is a consistent accounting of what it costs to distinguish something — and that accounting has a small number of provable properties.

The working name for the accounting is the **Universal Binary Principle** (UBP). "Binary" does not mean twoness; it means *difference as the first condition of distinction*. That reading is motivational: it does not enter any formula below, and nothing below depends on it.

1. The principle in one sentence

Distinction becomes possible when primitive difference is given structure, protection, embodiment, and read access — and each of those four has a price.

The severe form, with the prices named:

Where difference is not yet formed there is vacuum (price 0); where difference is formed but unlawful there is a syndrome (price up to $4Q$); where difference is encoded, protected and read there is measurement (price at least Q , and at least $8Q$ if it is to be protected).

Every number in that sentence is proved below.

2. Dual terminology, pinned down

Each term now names exactly one mathematical object.

Structural term	Symbol	Type	Operational meaning
Perfect space	0	pattern	no active coordinate
Zero vector	0	pattern	no disturbance, no information
Raw information	v	$\text{Fin } n \rightarrow \mathbb{Z}$	an integer pattern on $n = 24$ coordinates
Primitive difference "2"	Δ	$\mathbb{R} = 2$	the numerator of the read operator
Capacity / zone-share	$\mathbb{Z} \star$	$\mathbb{R} = 1/8$	cost of occupying a permitted zone
Body	24 coordinates	index set	the coordinate space
Loop-check (numeric)	Π	$\mathbb{R} = \pi$	the argument of the read operator
Loop-check (structural)	$\sigma(v)$	12 bits	the Golay syndrome
Not-quite-closed loop	$\sigma(v) \neq 0$	—	history, gap, syndrome
MOG	nearest-codeword reading	—	the grammar that turns a pattern into a lawful one
Golay	$[24, 12, 8]$ code	—	protection: minimum distance 8
Leech	Λ_{24}	—	embodiment: the 24-dimensional geometry
Observer / read quantum	Y	$\mathbb{R}$	$1/(\pi + 2/\pi) = 0.2646754\dots$
Activation quantum	Q	$\mathbb{R}$	$Y + 1/8 = 0.3896754\dots$

Structural term	Symbol	Type	Operational meaning
TAX	$\text{TAX}(\mathbf{v})$	$\mathbb{R}$	$\text{HW}(\mathbf{v}) \cdot \mathbf{Y} + \ \mathbf{v}\ ^2 / 8$
Coherence budget	B	$\mathbb{R} = 10$	the unit in which tax is measured
NRCI	$\text{NRCI}(\mathbf{v})$	$\mathbb{R}$	$B / (B + \text{TAX}(\mathbf{v}))$
CoherenceRegime	one of four	—	a band of NRCI , equivalently of TAX

Two symbols that were previously one. The original uses π both for the numeric loop-check inside $\mathbf{Y}$ and for the structural closure test. These are different objects and are now written $\Pi = \pi$ and $\sigma(\mathbf{v})$ respectively.

3. The sequence of steps

Each step supplies something the next one uses. The sequence is not claimed to be forced; it is claimed to be *coherent*, and every arrow in it is a real dependency in the formal development.

Step I — the perfect space [def + thm]

Begin with the vacuum: $\mathbf{v} = 0$. No active coordinate, no read event, no history.

[thm] $\text{TAX}(0) = 0$ and $\text{NRCI}(0) = 1$; and the vacuum is the **only** pattern with either property.
 $\text{tax_eq_zero_iff}, \text{nrci_eq_one_iff}$

Coherence is complete and knowledge is absent — and now in the strong sense: perfect coherence is not merely *compatible* with the absence of distinction, it is *equivalent* to it.

Step II — the first difference [stip]

The next condition is not a quantity but the possibility of difference. In the arithmetic it appears as a single stipulated constant:

$\Delta = 2$

[stip] $\Delta = 2$ enters only as the numerator of the read operator of Step IX. Any other positive value gives a consistent system with a different $\mathbf{Y}$. Nothing in this document tests it.

The motivational reading ("2 is a difference-state, not a number") is retained as motivation and is not used.

Step III — the disturbance $\mathbf{v}$ [def]

A pattern of difference is an integer vector $\mathbf{v}$. Two summaries of it are used throughout, and *only* these two:

$$\begin{aligned} \text{HW}(\mathbf{v}) &= \#\{i : v_i \neq 0\} && \text{the active distinctions} \\ \|\mathbf{v}\|^2 &= \sum_i (v_i)^2 && \text{the geometric extent} \end{aligned}$$

[thm] $\text{HW}(\mathbf{v}) \leq \|\mathbf{v}\|^2$, with equality exactly when every coordinate is $-1, 0$ or 1 . hw_le_normSq ,
 normSq_eq_hw_iff

At this stage $\mathbf{v}$ is not yet meaningful: it is detectable in principle and nothing more.

Step IV — capacity, zones, and the activation quantum [def + thm]

Placing $\mathbf{v}$ in a zoned body of 24 coordinates gives difference *location, limitation* and *addressability*. Entering a zone costs

$$Z\star = 1/8 \text{ [stip]}; \text{ reading a coordinate costs } \mathbb{Y}; \text{ so activating one coordinate costs}$$

$$Q = \mathbb{Y} + 1/8 = 0.3896754\dots$$

The original states this as an accounting identity. It is more than that:

[thm] For every nonzero pattern, $\text{TAX}(\mathbf{v}) \geq Q$; and $\text{TAX}(\mathbf{v}) = Q$ exactly when $\mathbf{v}$ has a single nonzero coordinate, equal to ± 1 . Q_le_tax , $tax_eq_Q_iff$

So Q earns the name *quantum* in the strict sense: nothing can be read for less, and the price Q is actually attained.

Step V — the loop and the gap [thm]

For a disturbance to be detected there must be a check, and the check must be able to *fail informatively*. A closed loop leaves no history; a not-quite-closed loop leaves a remainder, and the remainder is the record.

Operationally the loop-check is the syndrome $\sigma(\mathbf{v})$, and the image becomes three exact statements:

[thm] $\sigma(v) = 0 \iff v$ is lawful (a Golay codeword). $\text{loop_closes_iff_lawful}$ [thm] $\sigma(a \oplus b) = \sigma(a) \oplus \sigma(b)$: the record is additive. history_additive [thm] $\sigma(a) = \sigma(b) \iff a \oplus b$ is lawful. same_history_iff

The third is what entitles us to call the gap *the* history rather than a trace of it: the syndrome forgets precisely the lawful content and nothing else.

A measurement is therefore a state **plus** its record of closure failure — and the record is complete.

Step VI — MOG as distinction grammar [thm]

MOG is the rule that reads a raw pattern as a lawful one: choose the lawful pattern nearest to it. Its three operative facts:

[thm] every pattern lies within distance 4 of a lawful one (covering radius 4). $\text{golay_covering_radius}$ [thm] distinct lawful patterns differ in at least 8 coordinates (minimum distance 8), so a pattern with at most 3 errors has a unique reading. golay_min_dist [thm] at distance exactly 4 the reading is genuinely ambiguous: there are patterns with two lawful neighbours at distance 4. $\text{decoding_not_unique}$

So the grammar answers "is this closed / lawful / correctable / where is the gap", with the honest boundary: **correctable to 3, decodable to 4, ambiguous at 4.**

MOG does not create difference. It makes difference usable.

Step VII — Golay protection, and its price [thm]

Protection means that lawful distinctions survive disturbance: within the correction capacity, error can be healed and distinction survives; beyond it, distinction becomes fragile. The environment is not perfect everywhere — it is perfect within its correction capacity.

Protection is not free, and its price is exact:

*[thm] the cheapest unprotected distinction costs $Q = 0.3897$; the cheapest **protected** one costs $8Q = 3.1174$ (an octad). $\text{tax_eq_Q_iff, protection_costs_eight_quanta}$*

Protection multiplies the minimum cost of being read by exactly 8 — the minimum distance of the code. That factor is the whole content of "protection costs something".

Step VIII — Leech embodiment [def + thm]

The Leech lattice gives the protected code a body: 24 dimensions, high symmetry, dense relational structure, no cheap local mirrors. Observation therefore cannot rely on local reflection; it needs a loop, a history and an observer.

In the substrate's integer scaling every minimal (kissing-sphere) vector has $\|\mathbf{v}\|^2 = 32$, so its tax is $\text{HW} \cdot \mathbf{Y} + 4$ and the three shape classes are separated by weight alone:

Class	Shape	HW	TAX	NRCI	Regime
A	$(\mp 4^2, 0^{22})$	2	4.529351	0.688262	Coherent
B	$(\mp 2^8, 0^{16})$	8	6.117403	0.620447	Coherent
C	$(\mp 3, \pm 1^{23})$	24	10.352210	0.491347	Transitional

[thm] classes A and B are Coherent; class C is Transitional. `minimalVector_classAB_coherent,`
`minimalVector_classC_transitional`

This is the first genuinely falsifiable consequence of the accounting: the *deepest* shell of the geometry falls out of the Coherent band. Either density is expensive — a defensible reading — or the thresholds are mis-scaled (§8).

Step IX — the observer/read quantum Υ [def + stip + thm]

Υ is not an observer as a being. It is the price of a read: what it costs to turn a protected distinction into an observed event.

`$\Upsilon[\Pi] = 1/(\Pi + \Delta/\Pi)$` the read-cost operator [def]
 `$\Upsilon = \Upsilon[\pi]$  with  $\Delta = 2 = 1/(\pi + 2/\pi) = 0.2646754...$` [stip: $\Pi = \pi$]

The original calls Υ "the minimum cost of reading a reflected difference". That is the one claim in the study that does not survive checking, and the correction is worth stating plainly, because what remains is still exactly what the rest of the document needs:

[thm] the operator is capped: $\Upsilon[\Pi] \leq 1/(2\sqrt{\Delta})$ for every $\Pi > 0$, with equality only at $\Pi = \sqrt{\Delta}$.
`readCost_le_amgm` [thm] it has no positive lower bound: $\Upsilon[\Pi] \leq 1/\Pi$. `readCost_le_inv` [thm] hence $\Upsilon < 1/(2\sqrt{2}) = 0.3536$: π is not the extremal loop-check. `$\Upsilon_{1t\_amgm}$`

So the honest statement is:

γ is the observer/read quantum at loop-check π — the reciprocal of the loop cost $\pi + 2/\pi$. The operator bounds any read cost by $1/(2\sqrt{\Delta})$; the choice $\Pi = \pi$ is a stipulation of this investigation, not a consequence of it.

[open] Is there a functional of the loop that π extremises? The obvious candidate — $\gamma[\Pi]$ itself — is not one.

Without γ there is structure, code, geometry and difference, but no observation. That claim is untouched by the correction: it is about the role of γ , not about its value.

4. The activation quantum

$$Q = \gamma + z\star = \gamma + 1/8$$

To activate a coordinate the observer pays both the read cost and the zone-entry cost:

$$\text{activation} = \text{read} + \text{zone entry}$$

***[thm]** and Q is exactly the minimum tax of a nonzero pattern, attained by the single ± 1 activations. Q_{le_tax} , $tax_{eq_Q_iff}$*

5. TAX — the symmetry tax

$$\text{TAX}(\mathbf{v}) = \underbrace{\text{HW}(\mathbf{v}) \cdot \gamma}_{\text{read cost}} + \underbrace{\|\mathbf{v}\|^2 / 8}_{\text{embodiment cost}}$$

The first term is topological: the observer pays γ per active distinction. The second is geometric: the displacement of the pattern, normalised by the zone scale 8.

For a pattern whose coordinates are all $-1, 0$ or 1 we have $\|\mathbf{v}\|^2 = \text{HW}$, hence

$$\text{TAX} = \text{HW} \cdot (\gamma + 1/8) = \text{HW} \cdot Q$$

[thm] and this identity characterises exactly those patterns: $TAX(v) = HW(v) \cdot Q$ iff every coordinate lies in $\{-1, 0, 1\}$. *tax_eq_hw_mul_Q_iff*

Two consequences of the sharpened form:

- the identity is about $\{-1, 0, 1\}$ patterns, not about "binary" loosely understood; it holds for signed patterns and fails for any coordinate of size ≥ 2 ;
- it therefore holds on the **code** layer and fails on the **Leech** layer, where the analogue is $TAX = HW \cdot Y + 4$.

So, on the code layer: *the symmetry tax is the number of active distinctions times the activation quantum*. That is the result the original calls beautiful, now with its exact domain of validity attached.

A limitation to state openly.

[thm] on signed patterns TAX depends on HW alone, so a lawful codeword and a random error pattern of the same weight are taxed identically. *tax_eq_of_hw_eq*

The tax, as defined, cannot see lawfulness. §14 A repairs this.

6. NRCI — coherence after tax

$$NRCI(v) = B / (B + TAX(v)), \quad B = 10$$

[thm] $NRCI$ is a strictly decreasing function of the tax alone, with values in $(0, 1]$; it equals 1 only at the vacuum and is never 0. *coh_strictAnti, coh_pos, coh_le_one, nrci_eq_one_iff* **[thm]** the budget is a **pure scale**: $NRCI$ depends only on TAX/B . *nrciB_eq_coh*

The original proposes reading the budget as $B = Z + \Delta = 8 + 2 = 10$. That arithmetic is consistent, but the scaling theorem shows what its status is:

[stip] B fixes the unit in which tax is measured. The reading $B = Z + \Delta$ is an interpretation of that unit. Because $NRCI$ depends only on TAX/B , no measurement inside the system can confirm or refute the reading unless the regime thresholds are fixed independently.

This is not a weakness. It is a degree of freedom, and §8 uses it.

7. The zero vector

For $v = 0$: $HW = 0$, $\|v\|^2 = 0$, $TAX = 0$, $NRCI = 1$.

Perfect coherence, and zero information. Not a flaw but the central principle, now with an exact form:

[thm] with $info(v) := 1 - NRCI(v) = TAX/(B + TAX)$, we have $NRCI + info = 1$ identically, and $info = 0$ exactly at the vacuum. `coh_add_info, nrci_eq_one_iff`

To know something, coherence must be spent. Coherence and information are complementary fractions of one budget.

8. Coherence regimes

The four regimes are thresholds on $NRCI$. Because $NRCI$ decreases strictly in the tax, each threshold is a tax ceiling B/c

– B:

Regime	NRCI	TAX band	signed weight	reachable on 24 coordinates?
OnBit	≥ 0.8	$0 \dots 2.5$	$HW \leq 6$	yes
Coherent	≥ 0.5	$2.5 \dots 10$	$HW \leq 25$	yes
Transitional	≥ 0.3	$10 \dots 23.33$	$HW \leq 59$	no
Subcoherent	< 0.3	> 23.33	—	no

[thm] the four regimes are exactly these four bands. `regime_eq_onBit_iff`, `regime_eq_coherent_iff`, `regime_eq_transitional_iff`, `regime_eq_subcoherent_iff` [thm] on 24 coordinates a signed pattern has $TAX \leq 24Q = 9.3522 < 10$, so only OnBit and Coherent can occur. `signed24_regime` [thm] OnBit means exactly "at most six active distinctions". `signed_onBit_iff` [thm] every nonzero Golay codeword is Coherent — never OnBit. `golay_regime_coherent`

So, as calibrated with $B = 10$, the ladder is flat where it matters: the vacuum is OnBit and every protected distinction is Coherent. The table distinguishes nothing on the code layer.

The repair, using the freedom identified in §6. Take the cheapest protected distinction as the unit, $B = 8Q$. Then for a codeword of weight w

$$NRCI(w) = 8Q / (8Q + wQ) = 8 / (8 + w)$$

— the read cost cancels, leaving the weight measured against the minimum distance:

Weight	$NRCI = 8 / (8+w)$	Regime
0	1.000	OnBit
8	0.500	Coherent
12	0.400	Transitional
16	0.333	Transitional
24	0.250	Subcoherent

*[thm] $NRCI(w) = 8 / (8+w)$ on the calibrated budget, and the five weight classes then occupy all four regimes.
 $nrciB_calibrated, calibrated_regime_separates$*

All four regimes are used, the order is monotone in weight, and the octad — the minimum-weight protected distinction — sits exactly on the Coherent boundary at $NRCI = 1/2$. No definition changes; only the unit does.

9. MOG, Golay and Leech as one instrument

MOG	gives distinction grammar	reading is nearest-codeword choice (unique to 3, decodable to 4)
↓		
Golay	gives protection and correction	minimum distance 8; price 8Q
↓		
Leech	gives embodied geometry	$\ v\ ^2 = 32$ on the minimal shell
↓		
Y	pays the read	Q per activation
↓		
TAX, NRCI		the cost, and what remains

MOG reads, Golay protects, Leech embodies, Y observes — and each of the four now has a number attached: 3/4, 8, 32, Q.

10. What v is

v is not the possibility of difference; it is a pattern of difference — raw information as a disturbance of the perfect space. It becomes a measurement by passing through the instrument:

v → zoned body	location, limitation	[def]
→ loop-check $\sigma(v)$	syndrome / history	[thm: complete record]
→ MOG	lawful vs unlawful reading	[thm: unique to 3]
→ Golay	correctable distinction	[thm: price 8Q]
→ Leech	geometric distinction	[def]
→ Y	measurement	[thm: price $\geq Q$]

11. What Y is

As formula. $Y = 1 / (\pi + 2/\pi)$, the read operator $Y[\Pi] = 1 / (\Pi + \Delta/\Pi)$ at $\Delta = 2, \Pi = \pi$.

As operation. The observer/read quantum: the cost of one read.

As measurement principle. The per-distinction term of the tax. *Not* a minimum of the operator — the operator has no positive minimum (§ Step IX).

As activation component. $Q = Y + Z\star$, and Q is a minimum (tax_eq_Q_iff).

As "why". Without Y there is structure, code, geometry and difference, but no observation. Y is the principle by which the system becomes known to itself. This is a statement about the *role* of Y and it is unaffected by the correction to its extremal status.

12. Why the shape is convincing

Not because it is pretty, but because several independent requirements resolve into one accounting, and each link is checkable:

Requirement	Supplied by	Checkable content
measurement requires distinction	Δ	stipulated
distinction requires structure	MOG	unique reading to radius 3
structure requires protection	Golay	minimum distance 8
protection requires embodiment	Leech	$\ v\ ^2 = 32$ minimal shell
embodiment requires observation	Y	Q per activation
observation requires cost	TAX	$TAX \geq Q, = HW \cdot Q$ on signed patterns
cost requires accounting	NRCI	strictly decreasing, $(0, 1]$, scale-free in TAX/B

$\Delta \rightarrow v \rightarrow \text{MOG} \rightarrow \text{Golay} \rightarrow \text{Leech} \rightarrow Y \rightarrow \text{TAX} \rightarrow \text{NRCI}$

13. The complete map

PERFECT SPACE	TAX = 0, NRCI = 1	[thm: unique]
primitive difference $\Delta = 2$	[stip]	
disturbance v arises	HW, $\ v\ ^2$	[def]
zoned embodied body	24 coordinates, $z^\star = 1/8$	
loop-check $\sigma(v)$	$\sigma = 0 \iff$ lawful	[thm]
syndrome / history	complete record	[thm]
MOG reading	unique to 3	[thm]
Golay protection	$d = 8$, price 8Q	[thm]
Leech embodiment	$\ v\ ^2 = 32$	[def]
Observer / read Y	$Y = Y[\pi]$	[stip: $\Pi = \pi$]
activation $Q = Y + 1/8$	min tax	[thm]
TAX	HW $\cdot$ Q on signed	[thm]
NRCI	$B/(B+TAX)$	[thm]
CoherenceRegime	four tax bands	[thm]

14. Refinement horizon

A. MOG-aware TAX — resolved

The gap is real: the present tax cannot distinguish a lawful pattern from a random one of the same weight (tax_eq_of_hw_eq). A term that reads the syndrome is therefore necessary, and there is a canonical one — charge the pattern for the correction it needs:

```
syndromePenalty(v) = HW(leader(σ(v))) · Q
TAX_MOG(v)          = HW(v) · Q + syndromePenalty(v)
```

[thm] the penalty vanishes iff v is lawful; it never exceeds $4Q$ (the covering radius, priced in activation quanta); hence $TAX_MOG = TAX$ iff v is lawful. $syndromePenalty_eq_zero_iff$, $syndromePenalty_le$, $taxMOG_eq_bitTax_iff$

The original's "syndrome penalty" and "closure credit" are the same term seen from two sides: the credit is the absence of the penalty.

B. Shell-version NRCI — open, with the obstruction identified

A shell-aware coherence must not use $\|v\|^2$: on the Leech minimal shell that quantity is constant (32) and so carries no information. It must use HW , or the shape class, or the distance to the shell.

C. Codeword versus error pattern — resolved by A

D. Origin of the coherence budget — clarified

B is a scale ($nrciB_eq_coh$), so $B = Z + \Delta$ is untestable as it stands. The constructive alternative is to *derive* B rather than stipulate it: taking $B = 8Q$, the cheapest protected distinction, makes the budget a consequence of the code and makes all four regimes informative (§8).

15. What this system is not

It is not claimed that this is the only possible system; that this is what reality is; that Golay or Leech are ultimate physical structures; that the numbers are magic; or that Υ has been established as a physical constant. Two further disclaimers are added by this restatement:

- Υ is not derived; it is the read operator evaluated at the stipulated $\Pi = \pi$;
- the coherence budget and the four thresholds are stipulated together; only their ratio has meaning.

The claim that remains is narrow and defensible:

Given the stipulated constants and the chosen protected geometry, a coherent measurement accounting appears. It has an exact minimum ($\mathcal{Q}$), an exact price of protection ($8\mathcal{Q}$), a complete record of failure (the syndrome), and a coherence measure with the properties one would want. Whether it maps onto anything deeper remains open.

16. Final statement

The UBP, as seen here, is not a theory of everything. It is a dark-space instrument, and each part of it now has a number:

- it begins with perfect coherence — the unique zero-tax state;
- it allows difference — at a stipulated $\Delta = 2$;
- it gives difference form through ∇ — summarised by HW and $\|\nabla\|^2$;
- it gives ∇ zones and body — at $1/8$ per zone;
- it checks the loop through the syndrome — which records exactly the unlawful part;
- it encodes distinction through MOG — uniquely up to 3 errors;
- it protects distinction through Golay — at a price of $8\mathcal{Q}$;
- it embodies distinction through Leech — on a shell of norm 32;
- it reads distinction through Υ — at $\mathcal{Q}$ per activation, $\mathcal{Q}$ being the least possible;
- it prices distinction through TAX — $\text{HW} \cdot \mathcal{Q}$ on the code layer;
- it measures what remains through NRCI — a strictly decreasing, scale-free function of the tax.

And in this way the system sees itself just enough to know that it has been read — and now, to know what that reading cost.

17. The core in one breath

The vacuum is perfect coherence, and nothing else is. Difference disturbs it, at a price no smaller than $\mathcal{Q}$. The loop gives the disturbance a history, and the history is complete. MOG gives the history grammar, uniquely up to three errors. Golay gives the grammar protection, at eight quanta. Leech gives the protection a body, on a shell of norm thirty-two. Υ gives the body a reader, at the loop-check π we chose. TAX measures the cost of being read. NRCI tells how much coherence remains — in whatever unit we fix.